

Information Field Theory

Philosophy of Science and Epistemology

IFT-Philosophy v1.0

Juan Diego Vicente Gabancho

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The philosophical and epistemological foundation of Information Field Theory (IFT) is presented. Informational structural realism is proposed: physical reality is constituted by objective informational structures described by the field $\rho(x) > 0$, and our theories capture these structures in an approximate manner. The subject-object dichotomy is dissolved by reinterpreting the observer as an informational subsystem whose coupling with the environment is mediated by ρ . IFT is compared with scientific realism, structuralism, transcendental idealism, and phenomenology. It is argued that relational information is the fundamental ontological category, prior to matter, spacetime, and mind. Implications for the measurement problem in quantum mechanics and the status of non-observable theoretical entities are discussed.

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Chapter 1

Introduction: The Ontological Question

1.1 What is the world made of?

Physics has historically oscillated among various answers: matter (atomism), energy (thermodynamics), fields (electromagnetism), curved spacetime (general relativity), wave functions (quantum mechanics), strings (string theory). IFT proposes a unifying answer: the world is made of relational information, encoded in the field $\rho(x) > 0$.

1.2 Central Thesis

Theorem 1.2.1 (IFT Ontological Thesis). *Relational information, described mathematically by the field $\rho(x)$, is the fundamental ontological category. Matter, energy, spacetime, and mind are emergent manifestations of informational organization.*

Chapter 2

Informational Structural Realism

2.1 Definition

Definition 2.1.1 (Informational Structural Realism). 1. ***Realism:** There exists a reality independent of our mind.*

2. ***Structural:** What we know of that reality are its structures (relations, patterns, organization), not its intrinsic “substances”.*

3. ***Informational:** The fundamental structure of the universe is informational in character, captured by the field $\rho(x)$.*

2.2 Comparison with Standard Scientific Realism

Scientific realism (Putnam, Boyd) asserts that mature scientific theories are approximately true and that theoretical terms (electrons, quarks) refer to real entities. IFT accepts realism but specifies: what is real is not “particles” in themselves, but the informational structures that underlie them. An electron is a stable excitation (soliton) of the field ρ in a specific gauge sector; its “reality” is that of the informational structure that constitutes it.

2.3 Comparison with Structuralism (Worrall, Ladyman)

Scientific structuralism asserts that the content of scientific theories is purely structural (mathematical relations). IFT goes further: it specifies that the fundamental structure is of an informational type $(\rho, \Xi, S[\rho])$ and that this structure is ontologically primary, not merely epistemic.

2.4 Advantages of Informational Structural Realism

1. **Ontological unification:** A single type of fundamental entity (information) underlies all physical phenomena.
2. **Resistance to underdetermination:** Different empirically equivalent theories can capture the same underlying informational structure.
3. **Theoretical continuity:** Scientific progress consists in increasingly better capturing the structure ρ , not in “approaching truth” in an absolute sense.

Chapter 3

Dissolution of the Subject-Object Dichotomy

3.1 The Problem

Modern philosophy (Descartes, Kant) established a radical separation between the knowing subject and the known object. This dichotomy generates insoluble problems: how can the mind access the external world? Is knowledge a construction of the subject or a reflection of reality?

3.2 IFT Solution

IFT dissolves the dichotomy by reinterpreting both poles as coupled informational subsystems:

$$\rho_{\text{total}} = \rho_{\text{subject}} \otimes \rho_{\text{object}} + \text{correlation terms.} \quad (3.1)$$

“Knowledge” is the mutual information between the subject-subsystem and the object-subsystem:

$$I(\text{subject} : \text{object}) = S[\rho_{\text{subject}}] + S[\rho_{\text{object}}] - S[\rho_{\text{total}}]. \quad (3.2)$$

There is no “abyss” between subject and object: both are regions of the same informational field, and their interaction is a flow of information governed by the dynamics of ρ .

3.3 Implications

- Perception is a process of information transfer from the environment to the organism.
- Knowledge is not “internal representation of an external world”, but informational synchronization between subsystems.
- Consciousness (see IFT-Consciousness) is informational self-presence: the system models itself within its own ρ .

Chapter 4

Ontology of Relational Information

4.1 Information as a Fundamental Category

IFT proposes an ontological hierarchy:

1. **Fundamental level:** Relational information (ρ). It is not “made of” anything else; it is the ultimate constituent.
2. **Emergent level 1:** Geometry (spacetime). Emerges from $\tilde{g}_{\mu\nu}[\rho]$.
3. **Emergent level 2:** Matter and energy. Emerge from excitations and solitons of ρ .
4. **Emergent level 3:** Life and consciousness. Emerge from complex informational organization ($\Phi_{\text{IFT}} > \Phi_c$).

4.2 What is Information?

Information is not a “thing” in the traditional sense. It is a relational category: $\rho(x)$ specifies how the different points of the field relate to one another. There is no information without relation, and there is no relation that is not informational.

4.3 Comparison with Idealism

IFT is not idealism: $\rho(x)$ exists objectively, independently of any mind. But neither is it naive materialism: “matter” is an emergent manifestation of ρ , not its foundation.

4.4 Comparison with Phenomenology

Phenomenology (Husserl, Merleau-Ponty) emphasizes lived experience as the starting point. IFT is compatible: lived experience is the first-person perspective of a system with $\Phi_{\text{IFT}} > \Phi_c$. But IFT adds a third-person description (the field ρ) that complements the phenomenological description.

Chapter 5

Implications for the Measurement Problem in Quantum Mechanics

5.1 The Problem

How and when does the “collapse” of the wave function occur? What distinguishes a measurement from an ordinary physical interaction?

5.2 IFT Solution

In IFT, there is no collapse. Measurement is a process of informational flow between the measured system and the measurement apparatus:

$$\rho_{\text{total}}(t) = \rho_S(t) \otimes \rho_A(t) \xrightarrow{\text{interaction}} \rho_{SA}(t). \quad (5.1)$$

“Measurement” occurs when the mutual information $I(S : A)$ reaches a critical threshold: the apparatus becomes correlated with the system irreversibly (decoherence). What we call a “measurement result” is simply the final state of the apparatus, which is correlated with the system. There is no additional ingredient (collapse, observer’s consciousness) beyond the informational flow described by the dynamics of ρ .

Chapter 6

Status of Non-Observable Theoretical Entities

6.1 Do quarks, gluons, the Higgs field exist?

According to informational structural realism:

- **They exist** as stable informational structures (solitons, excitations of the field ρ in specific gauge sectors).
- **They do not exist** as “point particles” with intrinsic properties independent of the experimental context.
- Their reality is relational: a quark is what it is by virtue of its (informational) interactions with other entities, not because of an internal “essence”.

6.2 Non-Observable Entities vs. Theoretical Constructs

IFT draws a clear line: a theoretical entity is real if it corresponds to a stable informational structure ($\tau\Xi > K$, i.e., it persists temporally). If it is purely a construct without informational stability (e.g., luminiferous ether, phlogiston), it is not real.

Chapter 7

Comparative Table with Philosophical Positions

Position	External Reality	Fundamental Entities	Subject-Object	Knowledge
Scientific Realism	Yes	Particles, fields	Separated	Approximation
Structuralism	Yes	Structures	Separated	Structural
Idealism	No	Ideas, mind	Unity	Construction
Phenomenology	Yes (phenomena)	Experience	Intentionality	Embodied
IFT	Yes	ρ (relational info)	Coupled	Synchronization

Chapter 8

Final Declaration

IFT-Philosophy provides ontological and epistemological foundations for the IFT framework. Informational structural realism asserts that reality is constituted by relational information (ρ), that knowledge is informational synchronization between subsystems, and that the subject-object dichotomy dissolves in informational flow. This position unifies scientific realism, structuralism, and phenomenology under a single conceptual framework.

$$\boxed{\text{Reality} = \text{Informational structure } \rho(x)}.$$

$$\boxed{\text{Knowledge} = \text{Mutual information } I(\text{subject} : \text{object})}.$$

$$\boxed{\text{Matter, spacetime, mind} = \text{Emergent manifestations of } \rho}.$$

8.1 Affirmations

1. Relational information $\rho(x)$ is the fundamental ontological category.
2. Informational structural realism unifies realism, structuralism, and phenomenology.
3. The subject-object dichotomy dissolves in informational flow.
4. Quantum measurement is informational decoherence, not collapse.
5. Theoretical entities are real if they have informational stability ($\tau\Xi > K$).

8.2 Open Problems

1. Is ρ itself an “informational” entity, or is it a mathematical formalism representing something more fundamental? (Realism vs. instrumentalism debate.)
2. How is the correct ρ selected among multiple equivalent informational descriptions?
3. What is the ontological status of the field ρ prior to any observer? (Realism without an observer?)
4. Can IFT derive the laws of logic and mathematics from informational principles?